

Review

1. Limits

(a) Compute the following limits:

(a) $\lim_{x \rightarrow +\infty} \frac{\sqrt{3x+4x^8}}{3x^4-x}$, (without using L'Hopital)

(b) $\lim_{x \rightarrow 0^+} \frac{\sqrt{3x+4x^8}}{3x^4-x}$, (without using L'Hopital)

(c) $\lim_{x \rightarrow 0} 3 \ln(x+1) \cdot \cos \frac{1}{x}$

(b) Find all the values of $a \in \mathbb{R}$ for which the following limit exist. Then, for this values, compute the limit:

$$\lim_{x \rightarrow -2} \frac{\sqrt{x^2 + 2a - 1} - a}{x + 2}$$

(c) Prove, using the *definition of limit* that:

$$\lim_{x \rightarrow -3} (x^2 - x + 2) = 14$$

2. Continuity

(a) Determine for which parameters $a, b \in \mathbb{R}$ the following function continuous is:

$$f(x) = \begin{cases} \frac{\sqrt{x+3}-\sqrt{3x+1}}{\sqrt{x-1}} & \text{if } x > 1, \\ ax + b & \text{if } -2 \leq x \leq 1 \\ \frac{x^2+2x}{x^2+x-2} & \text{if } x < -2. \end{cases} \quad (1)$$

(b) Prove that there exists $c \in (1, e)$ such that $\frac{\ln c}{c} = \frac{1}{3}$.

3. Differentiability

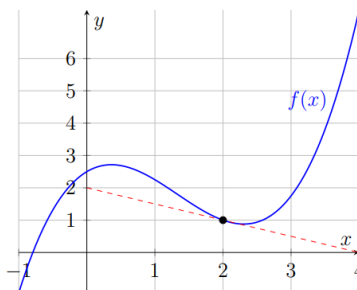
(a) Find the derivative of the following function *using the definition*.. State the domain of it and the domain of its derivative

$$G(t) = \frac{1-2t}{3+t}$$

(b) Find all $a, b, c \in \mathbb{R}$ such that the following function is differentiable:

$$f(x) = \begin{cases} \ln x & \text{if } x \geq 1, \\ a(x-1)^2 + b(x-1) + c & \text{if } x < 1 \end{cases} \quad (2)$$

(c) The line tangent to $f(x)$ at $x = 2$ is shown in the picture:



If $g(x) = x^2 + 1$ and $F = f \circ g$. Find $F'(1)$.

- (d) Find the equation of the tangent line to the ellipse using *implicit differentiation*

$$\frac{x^2}{75} + \frac{y^2}{6} = 1$$

at the point $P = (5, -2)$.

4. Applications of Derivative

- (a) Determine the local extremes, increasing/decreasing intervals, inflexion points, concavity intervals and sketch the graph of the following function:

$$f(x) = \frac{x^2 - 4}{x^2 - 3}$$

- (b) A number a is called a *fixed point* of a function f if $f(a) = a$. Prove that if $f'(x) \neq 1$ for all real numbers x , then f has at most one fixed point.
- (c) Find the following limit:

$$\lim_{x \rightarrow 1^+} \left(\frac{x}{x-1} - \frac{1}{\ln x} \right)$$

- (d) Find an equation of the line through the point $(3, 5)$ that cuts off the least area from the first quadrant.